

AZKA SHAUKAT

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PROFESSIONAL SUMMARY

Software Engineering graduate (CGPA: 3.86/4.00) with end-to-end experience building production-grade full-stack web applications integrated with practical AI/ML components — from a 6-model PyTorch deepfake detection ensemble to scikit-learn classification pipelines and locally-deployed LLMs. Architected a 7-microservice cybersecurity ecosystem with 400+ automated test cases, accessible via both a WhatsApp chatbot and a React web portal with real-time WebSocket streaming. Currently automating smart meter analytics at Pakistan's largest electricity distributor (LESCO, 6.9M consumers), replacing manual spreadsheet workflows with Python anomaly detection pipelines. Seeking an entry-level Full Stack AI/ML Engineer role to build impactful, user-facing intelligent systems.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, C++ (Core) C#, Java, PHP (Secondary)
Frontend	React 18/19, Next.js, Vite, Tailwind CSS, shadcn/ui, Zustand, Framer Motion, HTML5, CSS3, Bootstrap, jQuery
Backend & APIs	FastAPI, REST APIs, WebSocket, JWT Authentication
AI / ML	PyTorch, scikit-learn (Random Forest, ML Pipelines), Ollama (LLaMA 3.2, Mistral, LLaVA 7B), OpenCV, Pyzbar
Databases	PostgreSQL, MySQL, Redis (caching & sessions), SQLite, Pinecone (vector DB)
DevOps & Tools	Docker, Docker Compose, Git, GitHub, Nginx, VS Code
Game Development	Unity (2D), C# Scripting, Prefab System, Sprite Animations, Audio Integration
Software Engineering	Microservices Architecture, MVC/MVS Patterns, SDLC, Agile, UML Diagrams, Technical Documentation
Security	SHA-256 Hashing, HMAC Validation, HIBP k-Anonymization, JWT Dual-Token Auth, bcrypt, PBKDF2, GDPR Compliance

EXPERIENCE

IT Intern — AMI Cell, Project Implementation Unit, LESCO	Jul 2026 – Present
- Build a full-stack web application (Python, FastAPI, React) for the Advanced Metering Infrastructure (AMI) Cell, automating a manual, file-based data analysis process for smart electricity meters across a network serving 6.9 million consumers. - Operate the Meter Data Management (MDM) portal for Automated Meter Reading (AMR) meters, executing meter profile creations, configuration updations, and firmware management across the utility's AMI infrastructure. - Develop Python automation scripts using Pandas to replace manual Excel workflows — automating data cleaning, sorting, validation, formula calculation, and reporting for 15,000+ row datasets, reducing audit turnaround from days to minutes. - Design backend data pipelines to parse, structure, and validate uploaded meter and usage data files, establishing the technical groundwork for future electricity-theft detection through automated consumption anomaly analysis.	
Online Mathematics Tutor (Freelance)	Jan 2025 – Jul 2026
- Delivered structured online mathematics tutoring to a matriculation-level student via Zoom over approximately 1.5 years, maintaining consistent weekly lesson schedules and achieving measurable improvement in student academic performance. - Designed customized lesson plans covering algebra, geometry, and arithmetic, adapting teaching methods to the student's learning pace and reinforcing concept clarity through interactive problem-solving sessions and digital whiteboard demonstrations.	

EDUCATION

BS Software Engineering · Lahore Garrison University	2022 – 2026 3.86 / 4
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Software Architecture, OOP, Operating Systems, Computer Networks, Web Engineering	
Intermediate (Pre-Engineering) · Aspire Group of Colleges, Lahore	2020 – 2022 81.27%
Matriculation · Fazaia Intermediate College, Lahore	2018 – 2020 92.54%

PROJECTS

Aegis AI — Unified Cybersecurity Platform (Final Year Project)	github.com/AzkaShaukat/aegis-ai
FastAPI · React 18 · TypeScript · PyTorch · scikit-learn · Docker · PostgreSQL · Redis · Ollama · Pinecone	
Architected an enterprise-grade cybersecurity platform with 7 orchestrated microservices accessible via both a WhatsApp chatbot (Meta Cloud API) and a React web portal with real-time WebSocket streaming, backed by 400+ automated test cases.	

- Developed a 6-model ensemble Deepfake Detector (EfficientNet-B4, ViT-B/16, FrequencyCNN, Spatial CNN, Temporal Transformer, SRM-CNN) achieving AUC up to 0.97 on 140,000+ samples, featuring a 5-rule voting algorithm and Grad-CAM heatmaps.
- Engineered a 35-feature ML phishing classifier with Random Forest, reducing scan latency from 3 min to ~60 sec via async concurrency across 5 threat feeds (VirusTotal, URLScan, Google Safe Browsing), with dynamic XAI feature importance breakdowns.
- Built a privacy-preserving 5-tier credential validator with HIBP k-anonymity (SHA-1 prefixing), 40+ API key pattern scanners, 600+ disposable domain blacklist, and a GDPR-compliant Redis cache architecture with zero raw-credential storage.
- Integrated a local Ollama LLM (LLaMA 3.2) to dynamically translate raw JSON threat intelligence into actionable, plain-English Explainable AI (XAI) guidance following a strict "Reason → Verdict → Action" response framework.

LESCO Meter Data Analyzer

theft-detection-frontend.netlify.app

React 19 · FastAPI · Pandas · Recharts · Openpyxl · Python Threading

- Developed a full-stack audit application replacing days of manual spreadsheet inspection with automated multi-sheet parsing of messy vendor Excel exports, featuring dynamic MSN header detection and Multiplication Factor auto-scaling.
- Engineered a 14-day trailing average anomaly detection algorithm flagging $\geq 50\%$ consumption drops, and built a gap-outage correlation engine that classifies missing data intervals to differentiate power outages from potential electricity theft.
- Implemented an in-memory asynchronous job queue with server-side pagination for 15,000+ row datasets and interactive Recharts visualizations on an analytical dashboard with multi-sheet formatted Excel report export.

GPA Genie — Academic Calculator Web App

gpagenie.vercel.app

HTML5 · CSS3 · Bootstrap 3.3.7 · Vanilla JavaScript · jQuery · LocalStorage

- Designed and deployed a responsive 12-page web application enabling university students to calculate SGPA, CGPA, grade conversions, and percentage conversions with a mobile-first interface and zero-build-step architecture.
- Implemented a zero-latency client-side search engine for rapid navigation between calculators, with simulated authentication workflows and structured content pages.

Desktop Applications — GPA Genie & Library Management System

github.com/AzkaShaukat/GPAGenie_DesktopApp

Python · Tkinter · CustomTkinter · MySQL · Matplotlib · bcrypt · Pytest

- Built two database-connected desktop applications adhering strictly to the MVC design pattern with direct MySQL connectivity, parameterized SQL queries for injection prevention, and custom-styled Tkinter graphical interfaces.
- GPA Genie Desktop: Developed 4 academic calculators with dynamically embedded Matplotlib visualizations (pie, bar, donut charts), role-based PBKDF2-SHA256 authentication, and an integrated blog CMS with full CRUD functionality.
- Library Management System: Engineered a transactional book lending engine with ACID-compliant SQL transactions (commit/rollback), automated 14-day loan lifecycle, \$5.00/day overdue fine calculation, and comprehensive inventory tracking.

Mind Maze — 2D Puzzle Game (Fun Project)

Unity (2D) · C# Scripting · Prefab System · Sprite Animation · Audio Integration

- Designed and developed a multi-level 2D puzzle game in Unity, utilizing the Prefab system for modular, reusable level progression, and implementing custom C# scripts for collision handling, player movement, and score tracking.
- Integrated sprite-based animations, background music, and audio effects to create an immersive, responsive player experience.

Personal Portfolio Website

my-portfolio-rho-snowy-69.vercel.app

Next.js (App Router) · React 19 · TypeScript · Tailwind CSS · shadcn/ui · Framer Motion

- Built a modern, performant developer portfolio featuring dynamic data-driven project routing, smooth Framer Motion page transitions, and responsive, accessible UI components built with Tailwind CSS and shadcn/ui.

CERTIFICATIONS & AWARDS

Python For Data Science - Real Time Experience — Udemy (Data Science Lovers)

May, 2025

Credential: UC-830c4e89-3057-4664-aa23-3ada868d6381

English for the Professional World - Certified ESL Training — Udemy (Bespoke Learning Solutions)

May, 2025

Credential: UC-c0fc0e8a-4b40-4ee0-9a79-d1d85e4cd296

High Achiever Certificate (awarded 5 times) — Lahore Garrison University

2022 – 2026

Academic Excellence in BS Software Engineering

LANGUAGES

Urdu: Native

English: Professional Working Proficiency (Certified ESL Training)